

Original Articles: Vegetarian Compared with Meat Dietary Protein Source and Phosphorus Homeostasis in Chronic Kidney Disease



Summary Background and objectives Patients with advanced chronic kidney disease (CKD) are in positive phosphorus balance, but phosphorus levels are maintained in the normal range through phosphaturia induced by increases in fibroblast growth factor-23 (FGF23) and parathyroid hormone (PTH). This provides the rationale for recommendations to restrict dietary phosphate intake to 800 mg/d. However, the protein source of the phosphate may also be important. **Design, setting, participants, & measurements** We conducted a crossover trial in nine patients with a mean estimated GFR of 32 ml/min to directly compare vegetarian and meat diets with equivalent nutrients prepared by clinical research staff. During the last 24 hours of each 7-day diet period, subjects were hospitalized in a research center and urine and blood were frequently monitored. **Results** The results indicated that 1 week of a vegetarian diet led to lower serum phosphorus levels and decreased FGF23 levels. The inpatient stay demonstrated similar diurnal variation for blood phosphorus, calcium, PTH, and urine fractional excretion of phosphorus but significant differences between the vegetarian and meat diets. Finally, the 24-hour fractional excretion of phosphorus was highly correlated to a 2-hour fasting urine collection for the vegetarian diet but not the meat diet. **Conclusions** In summary, this study demonstrates that the source of protein has a significant effect on phosphorus homeostasis in patients with CKD. Therefore, dietary counseling of patients with CKD must include information on not only the amount of phosphate but also the source of protein from which the phosphate derives.